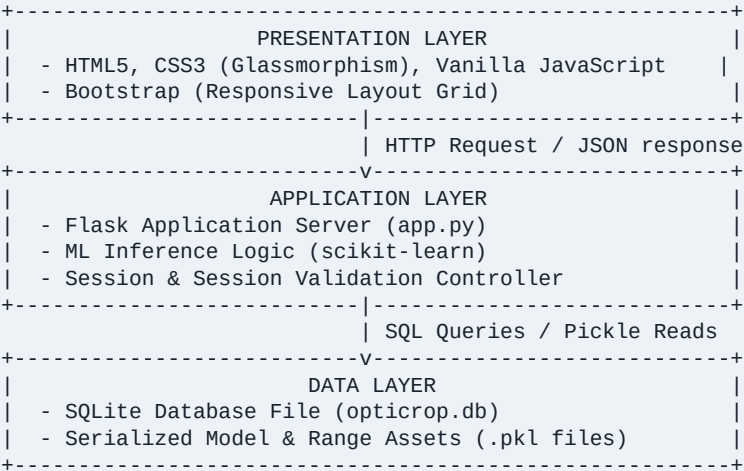


OPTI-CROP PROJECT DELIVERABLE

Solution Architecture

1. Three-Tier Architectural Layout

Opti-Crop is designed as a classic 3-tier web application, securing modularity, ease of testing, and clean separation of concerns.



2. Database Schema Relationships

The SQLite database (**opticrop.db**) maintains a structured schema with 7 interrelated tables to enable logging:

- **users:** Holds user profiles, hashed passwords, and access roles (Farmer vs Researcher).
- **soil_data:** Stores input soil properties, linked to the **users** table via foreign keys.
- **crops:** Stores optimal ph, season, water requirements, and crop categorization.
- **datasets & ml_models:** Stores model tracking information (model accuracy and datasets used).
- **predictions:** Links **soil_data**, **crops**, and **ml_models** with a timestamp and confidence score.
- **reports:** Stores generated summaries and chemical/organic recommendation text.